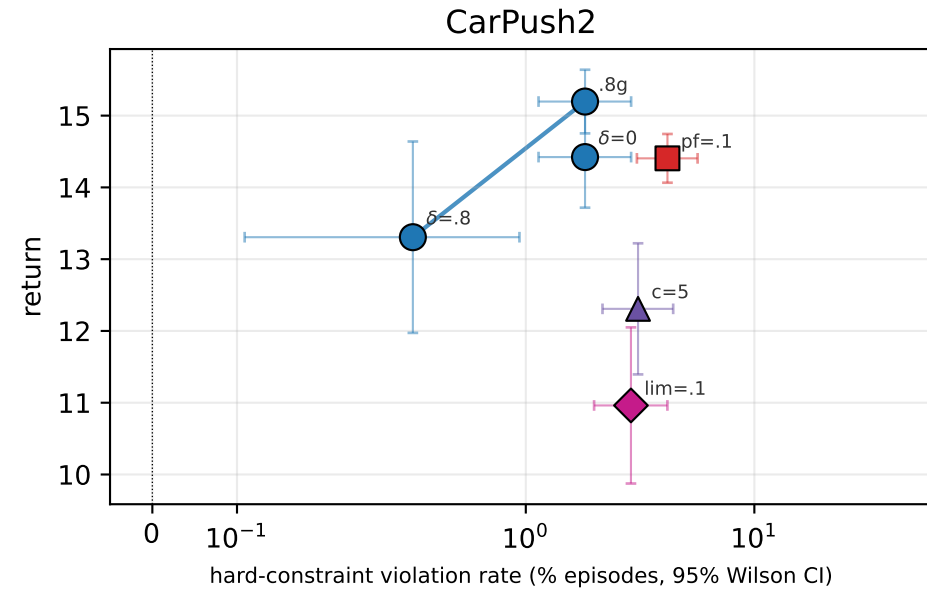
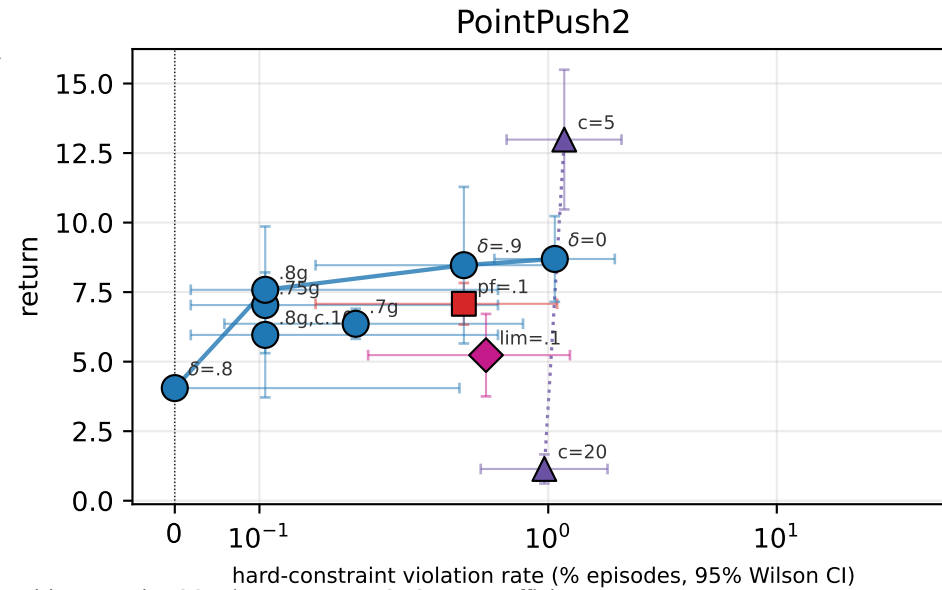
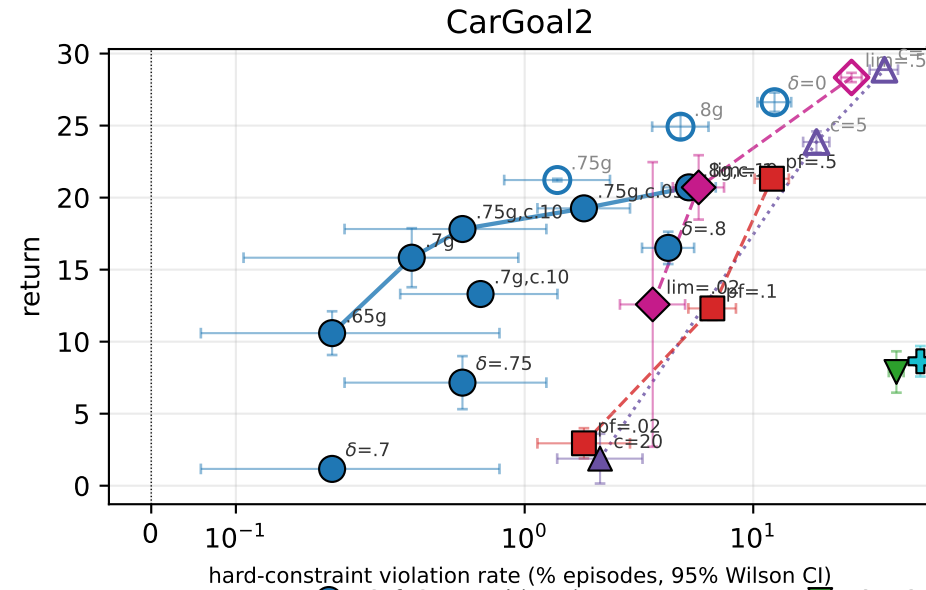
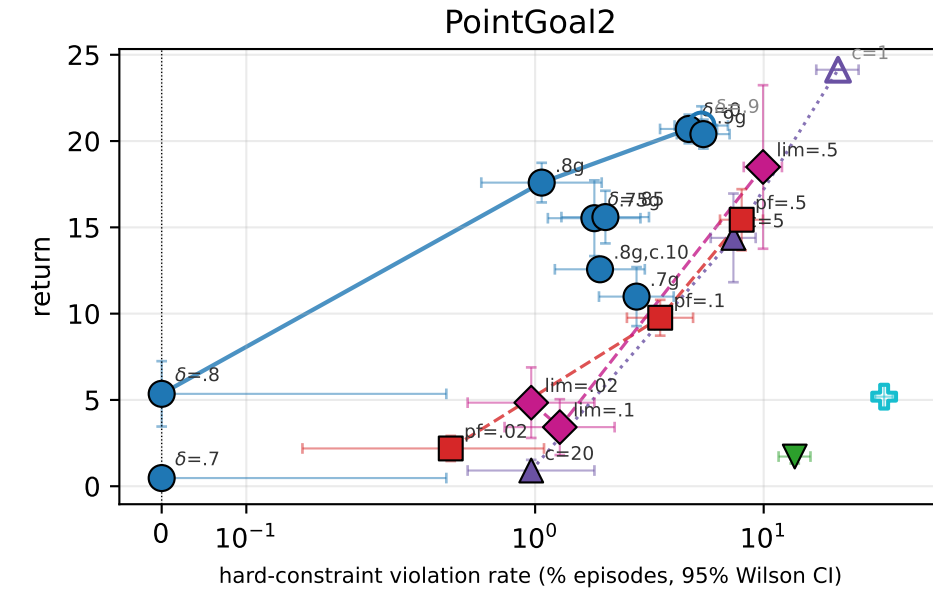


Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



- SafeOT-typed (ours)
- FDPI (constraint-based)

-  SAC (constrained least squares)
-  SAC-Lagrangian (penalty)
-  SAC + fixed penalty (penalty)

- ▲ SAC + fixed penalty (penalty-based)

- SDAC (multi-constraint SOTA)
- SRCPO (multi-constraint SOTA)

- typed Pareto frontier (soft ≤ 2)

constraint violation rate (% episodes, 95% Wilson CI)

- SAC-Pen coefficient sweep

FDRL is sweep

--- FDPI p_f sweep

- SAC-Lag cost-limit sweep